

Vapour pressures and excess Gibbs energy of propane-2-one + chloroalkanes binary mixtures

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Received 21 August 1997; accepted 17 December 1997

Abstract

Total vapour pressures have been measured for binary mixtures of propane-2-one with 1-chloropentane (I), 1,3-dichloropropane (II), and 1,1,1-trichloroethane (III), at 298.15, 303.15, 308.15 and 313.15 K, by a static method. The data were reduced using Barker’s method and different expressions for excess Gibbs energy were tested. The results evidenced positive (for (I) and (III) mixtures) and negative (for (II) mixture) deviations from ideality. © 1998 Elsevier Science B.V. All rights reserved.

Keywords: Vapour–liquid equilibria; Data; Gibbs energy; Vapour pressure; Mixtures; Propane-2-one; Chloroalkanes

1. Introduction

This paper is a part of a research project on vapour–liquid equilibria (VLE) in ketones + chloroalkanes mixtures and deals with isothermal VLE measurements of propane-2-one + 1-chloropentane, + 1,3-dichloropropane and + 1,1,1-trichloroethane mixtures, for which no such experimental data are available [1,2].

This studies will be continued with the investigation on other ketones + chloroalkanes mixtures and the results will be used for estimating interaction parameters for group contribution method. To correlate the experimental VLE data, different expressions for G^E were tested.

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2. Experimental

2.1. Materials

All materials were commercial products from E. Merck AG (Darmstadt and Schuchardt) of the first grade purity. The purity of substances checked by gas chromatography was not less than 99.8 mole%. Evidence of chemical purity was also obtained by comparison of refractive indices, $n_D^{298.15\text{ K}}$ and densities, $\rho^{298.15\text{ K}}$ with the literature values, in Table 1.

The liquids were dried and stored over 4A molecular sieves and used without further purification.

2.2. Apparatus and procedure

The vapour pressure measurements of pure compounds and binary mixtures were carried out by a static method. Use has been made of an isotenoscope described in detail elsewhere [3]. It is similar to that used by Surovy et al., [4]. Part of the equilibrium apparatus including a mercury filled U-tube, as a null manometer is placed in a fixed thermostated jacket; the equilibrium cell is equipped with a fitting-on thermostating mantle. The temperature of this mantle is the accurate equilibrium temperature, while the temperature in the fixed jacket is maintained 1 to 2 K higher, to prevent the partial condensation of vapours, in the upper part of the apparatus.

After thermostating to $\pm 10^{-2}$ K the equilibrium cell at the required temperature, the pressure exerted by the vapour is equilibrated in the null manometer against a pressure of air. In turn, the air pressure is measured using an external mercury manometer connected to the isotenoscope.

The manometer readings were performed with a Griffin and George (London, UK) Type 4214 cathetometer to ± 0.01 mm, and the pressure reproducibility was estimated to be better than 20 Pa. The measured values of the vapour pressure were corrected for mercury temperature and local gravity, but the hydrostatic pressure of the air was neglected.

The temperature of the liquid sample was measured with an accuracy of ± 0.01 K by means of a set of Beckmann thermometers, previously checked.

Samples with a volume of about 10 cm^3 of known composition were prepared by weight (± 0.1 mg) and thoroughly degassed into the equilibrium cell; the degassing procedure of alternate freezing, pumping to high vacuum and thawing, was similar to that used by Ronc and Ratcliff [5] and Young et

Table 1
Physical properties of pure compounds

Compound	$n_D^{298.15\text{ K}}$		$\rho^{298.15\text{ K}} (\text{kg/m}^3)$	
	This work	Literature	This work	Literature
Propane-2-one	1.3579	1.3561 ^a	784.4	784.2 ^b
1-Chloropentane	1.4101	1.4099 ^a	879.8	876.9 ^a
1,3-Dichloropropane	1.4459	1.4487 ^c	1180.5	1180.2 ^a
1,1,1-Trichloroethane	1.4350	1.4358 ^a	1331.3	1330.3 ^a

^a CRC—Handbook of Chemistry and Physics, 73rd Edition, 1992–1993.

^b J. Brown and F. Smith, Austral. J. Chem., (1957), 10, 423.

^c Value at 293.15 K.

al., [6]. (6–8 cycles were sufficient for the samples degassing). The solutions in the equilibrium cell were stirred with a magnetic stirrer and equilibrium was obtained within 1 h.

The final composition of the mixtures was determined refractometrically at 298.15 K, after each measurement of vapour pressure, by means of an Abbe refractometer ($n_D^{298.15\text{ K}} = \pm 0.0002$).

Several mixtures of known composition were used to construct the calibration curve.

Since a plot of refractive index of mixture, n , against mole fraction x was not accurate enough, n was fitted by a least-square method, to the excess function Y , as used by Philippe [7]:

$$Y = A_0 + A_1 X_{\text{ad}} + A_2 X_{\text{ad}}^2 + A_3 X_{\text{ad}}^3 + \dots \quad (1)$$

where: $X_{\text{ad}} = (n - n_1)/(n_2 - n_1)$, n_1 and n_2 are refractive indices of pure compounds.

The mole fraction x was obtained through:

$$Y = x - X_{\text{ad}} \quad (2)$$

For each of the studied mixtures the coefficients of Eq. (1) are specified as follows: (I) $A_0 = 0.981845$; $A_1 = -1.712324$; $A_2 = -0.267679$; (II) $A_0 = 0.987825$; $A_1 = -1.699067$; $A_2 = -0.286638$; (III) $A_0 = 1.00145$; $A_1 = -1.831274$; $A_2 = -0.005832$; $A_3 = -0.165349$. The estimated uncertainty in x is $\sigma(x) = 0.001$ mole fraction.

Table 2

Vapour pressure P , second virial coefficients B_{aa} and liquid molar volumes V^0 for pure compounds at working temperatures

Compound	P/kPa			$-B_{\text{aa}} (\text{cm}^3/\text{mol})$	$V^0 (\text{cm}^3/\text{mol})$
		This work	Literature		
Propane-2-one	T_1	30.74	30.82 ^a	1992	74.04
	T_2	37.85	38.05	1858	74.60
	T_3	46.28	46.60	1737	75.17
	T_4	56.24	56.65	1602	75.76
1-Chloropentane	T_1	4.25	4.02 ^b	3133	123.24
	T_2	5.43	5.19	2937	123.95
	T_3	6.89	6.64	2761	124.68
	T_4	8.65	8.42	2602	125.42
1,3-Dichloropropane	T_1	2.33	2.24 ^c	3195	96.80
	T_2	3.01	2.94	2983	97.32
	T_3	3.86	3.82	2792	97.85
	T_4	4.91	4.92	2622	98.39
1,1,1-Trichloroethane	T_1	16.62	16.49 ^d	1681	124.63
	T_2	20.43	20.49	1592	125.38
	T_3	24.97	25.25	1511	126.16
	T_4	30.36	30.86	1437	126.95

$T_1 = 298.15\text{ K}$; $T_2 = 303.15\text{ K}$; $T_3 = 308.15\text{ K}$; $T_4 = 313.15\text{ K}$.

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Table 3

Experimental and calculated VLE data in propane-2-one + 1-chloropentane binary mixture

x_1	y_1 calc	P_{exp} (kPa)	P_{calc} (kPa)	$P_e - P_c$ (kPa)	γ_1	γ_2	G^E (J/mol)
$T = 298.15$ K							
0.0000	0.0000	4.24	4.24	0.00	1.8121	1.0000	0
0.0647	0.4525	7.31	7.30	0.01	1.6930	1.0023	88
0.0730	0.4825	7.74	7.66	0.08	1.6791	1.0029	100
0.2362	0.7546	13.66	13.78	−0.12	1.4529	1.0297	274
0.3194	0.8101	16.25	16.32	−0.07	1.3632	1.0553	336
0.4476	0.8643	20.01	19.69	0.32	1.2478	1.1152	395
0.5995	0.9065	22.76	23.01	−0.25	1.1392	1.2336	402
0.6818	0.9249	24.58	24.62	−0.04	1.0919	1.3307	374
0.7843	0.9465	26.48	26.53	−0.05	1.0450	1.5020	303
0.8700	0.9653	28.33	28.13	0.20	1.0174	1.7089	210
0.9569	0.9872	30.10	29.84	0.26	1.0020	2.0096	79
1.0000	1.0000	30.74	30.74	0.00	1.0000	2.2073	0
$T = 303.15$ K							
0.0000	0.0000	5.43	5.43	0.00	1.8443	1.0000	0
0.0647	0.4425	9.18	9.18	0.00	1.6953	1.0028	92
0.0730	0.4719	9.59	9.62	−0.03	1.6785	1.0035	103
0.2362	0.7424	16.81	16.88	−0.07	1.4251	1.0335	274
0.3194	0.7995	20.04	19.97	0.07	1.3346	1.0599	332
0.4476	0.8568	23.94	23.95	−0.01	1.2252	1.1179	384
0.5995	0.9024	28.03	28.12	−0.09	1.1268	1.2264	386
0.6818	0.9223	30.13	30.15	−0.02	1.0843	1.3135	358
0.7843	0.9454	32.47	32.57	−0.10	1.0419	1.4669	289
0.8700	0.9650	34.92	34.60	0.32	1.0164	1.6528	200
0.9569	0.9872	36.92	36.73	0.19	1.0020	1.9252	76
1.0000	1.0000	37.84	37.84	0.00	1.0000	2.1057	0
$T = 308.15$ K							
0.0000	0.0000	6.88	6.88	0.00	1.7558	1.0000	0
0.0647	0.4250	11.20	11.28	−0.08	1.6425	1.0022	88
0.0730	0.4547	11.82	11.80	0.02	1.6293	1.0028	98
0.2362	0.7338	20.65	20.66	−0.01	1.4131	1.0292	266
0.3194	0.7930	24.35	24.35	0.00	1.3279	1.0541	324
0.4476	0.8519	29.49	29.29	0.20	1.2201	1.1113	377
0.5995	0.8987	33.91	34.28	−0.40	1.1212	1.2201	380
0.6818	0.9194	36.82	36.76	0.03	1.0792	1.3062	351
0.7843	0.9435	39.77	39.73	0.01	1.0384	1.4530	282
0.8700	0.9641	42.60	42.23	0.33	1.0147	1.6236	194
0.9569	0.9871	45.32	44.87	0.42	1.0017	1.8619	73
1.0000	1.0000	46.24	46.24	0.00	1.0000	2.0135	0
$T = 313.15$ K							
0.0000	0.0000	8.65	8.65	0.00	1.7211	1.0000	0
0.0647	0.4126	13.78	13.88	−0.10	1.6179	1.0021	86
0.0730	0.4422	14.59	14.51	0.08	1.6055	1.0026	96
0.2362	0.7245	25.10	25.11	−0.01	1.3974	1.0285	262
0.3194	0.7851	29.51	29.52	−0.01	1.3128	1.0535	319

Table 3 (continued)

x_1	y_1 calc	P_{exp} (kPa)	P_{calc} (kPa)	$P_e - P_c$ (kPa)	γ_1	γ_2	G^E (J/mol)
$T = 313.15$ K							
0.4476	0.8456	35.49	35.41	0.08	1.2055	1.1111	369
0.5995	0.8944	41.26	41.41	-0.15	1.1094	1.2177	367
0.6818	0.9163	44.44	44.43	0.01	1.0700	1.2989	366
0.8700	0.9637	51.36	51.23	0.13	1.0123	1.5769	182
0.9569	0.9873	54.68	54.53	0.15	1.0014	1.7687	67
1.0000	1.0000	56.23	56.23	0.00	1.0000	1.8849	0

The static apparatus was checked by T , P , x measurements with the reference mixture benzene + cyclohexane at 303.15 and 313.15 K. Our experimental data compared with those of Zielkiewicz [8] evidenced that for a deviation of 0.03 kPa of vapour pressure at $x = 0.5$, a deviation of 2% in GE at the value of 314 J/mol was obtained.

The VLE data and the thermodynamic quantities derived, proved the adequate operation of the apparatus.

3. Results and discussions

Table 2 lists the vapour pressures of pure substances measured by us at the working temperatures, compared with values obtained from Antoine coefficients, reported by other authors; the calculated liquid molar volumes and second virial coefficients are also presented.

Tables 3–5 present the experimental data and the calculated ones: mole fraction of vapour phase, y , the activity coefficients, γ_1 and γ_2 and the excess Gibbs energy G^E , obtained by means of

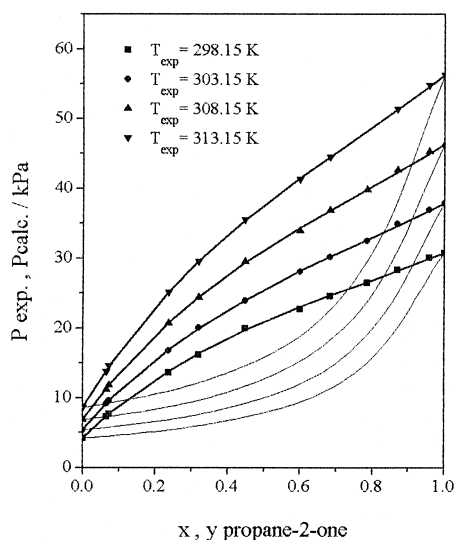


Fig. 1. Experimental and calculated isotherms for propane-2-one + 1-chloropentane mixture.

Table 4

Experimental and calculated VLE data in propane-2-one + 1,3-dichloropropane binary mixture

x_1	y_1 calc	P_{exp} (kPa)	P_{calc} (kPa)	$P_e - P_c$ (kPa)	γ_1	γ_2	G^E (J/mol)
$T = 298.15$ K							
0.0000	0.0000	2.33	2.33	0.00	1.0683	1.0000	0
0.0591	0.4555	4.00	4.07	−0.07	1.0317	1.0010	7
0.1565	0.6971	6.80	6.79	0.01	0.9968	1.0050	9
0.2768	0.8173	10.17	10.08	0.09	0.9820	1.0088	3
0.3582	0.8640	12.32	12.36	−0.04	0.9823	1.0085	−3
0.4420	0.8982	14.72	14.76	−0.04	0.9869	1.0053	−7
0.5214	0.9226	17.02	17.07	−0.05	0.9927	0.9998	−10
0.5999	0.9416	19.41	19.38	0.03	0.9980	0.9931	−10
0.6730	0.9560	21.57	21.52	0.05	1.0014	0.9872	−8
0.7467	0.9683	23.58	23.64	−0.06	1.0030	0.9835	−5
0.9339	0.9926	29.05	28.91	0.14	1.0007	1.0013	−2
1.0000	1.0000	30.74	30.74	0.00	1.0000	1.0250	0
$T = 303.15$ K							
0.0000	0.0000	3.01	3.01	0.00	1.0508	1.0000	0
0.0591	0.4406	5.10	5.12	−0.02	1.0202	1.0009	5
0.1565	0.6848	8.52	8.44	0.08	0.9906	1.0042	5
0.2768	0.8086	12.41	12.48	−0.07	0.9781	1.0075	−2
0.3582	0.8568	15.19	15.26	−0.07	0.9785	1.0072	−8
0.4420	0.8922	18.25	18.19	0.06	0.9829	1.0041	−13
0.5214	0.9177	21.08	21.01	0.07	0.9885	0.9988	−17
0.5999	0.9377	23.89	23.83	0.06	0.9938	0.9920	−17
0.6730	0.9531	26.29	26.44	−0.15	0.9977	0.9852	−16
0.7467	0.9660	29.07	29.07	0.00	1.0000	0.9794	−13
0.8414	0.9802	32.37	32.39	−0.02	1.0011	0.9764	−7
0.9339	0.9921	35.79	35.58	0.21	1.0004	0.9824	−2
1.0000	1.0000	37.84	37.84	0.00	1.0000	0.9953	0
$T = 308.15$ K							
0.0000	0.0000	3.86	3.86	0.00	1.0336	1.0000	0
0.0591	0.4261	6.43	6.41	0.02	1.0101	1.0007	3
0.1565	0.6731	10.44	10.46	−0.02	0.9880	1.0032	2
0.2768	0.8003	15.42	15.42	0.00	0.9798	1.0053	−5
0.3582	0.8499	18.93	18.84	0.09	0.9813	1.0045	−10
0.4420	0.8868	22.24	22.41	−0.17	0.9858	1.0014	−14
0.5214	0.9130	25.90	25.85	0.05	0.9909	0.9965	−16
0.5999	0.9339	29.31	29.27	0.04	0.9956	0.9906	−16
0.6730	0.9498	32.52	32.46	0.06	0.9989	0.9849	−15
0.7467	0.9636	35.60	35.63	−0.03	1.0008	0.9805	−11
0.8414	0.9786	39.53	39.65	−0.12	1.0012	0.9790	−6
0.9339	0.9914	43.62	43.53	0.09	1.0004	0.9861	−1
1.0000	1.0000	46.27	46.27	0.00	1.0000	0.9991	0
$T = 313.15$ K							
0.0000	0.0000	4.91	4.91	0.00	0.9871	1.0000	0
0.0591	0.4080	7.89	7.91	−0.02	0.9825	1.0001	−2
0.1565	0.6601	12.85	12.81	0.04	0.9778	1.0007	−8

Table 4 (continued)

x_1	y_1 calc	P_{exp} (kPa)	P_{calc} (kPa)	$P_e - P_c$ (kPa)	γ_1	γ_2	G^E (J/mol)
$T = 313.15$ K							
0.2768	0.7915	18.76	18.84	−0.08	0.9762	1.0011	−15
0.3582	0.8424	23.03	22.96	0.07	0.9772	1.0006	−21
0.4420	0.8801	27.24	27.22	0.02	0.9796	0.9989	−25
0.5214	0.9074	31.22	31.30	−0.08	0.9827	0.9960	−29
0.5999	0.9291	35.42	35.37	0.05	0.9863	0.9913	−31
0.6730	0.9461	39.13	39.19	−0.06	0.9899	0.9850	−31
0.7467	0.9609	43.15	43.06	0.09	0.9933	0.9767	−29
0.8414	0.9773	47.99	48.02	−0.03	0.9971	0.9625	−22
0.9339	0.9911	52.75	52.84	−0.09	0.9994	0.9443	−11
1.0000	1.0000	56.23	56.23	0.00	1.0000	0.9287	0

Table 5

Experimental and calculated VLE data in propane-2-one + 1,1,1-trichloroethane binary mixture

x_1	y_1 calc	P_{exp} (kPa)	P_{calc} (kPa)	$P_e - P_c$ (kPa)	γ_1	γ_2	G^E (J/mol)
$T = 298.15$ K							
0.0000	0.0000	16.62	16.62	0.00	1.5073	1.0000	0
0.0562	0.1364	18.36	18.20	0.16	1.4502	1.0011	54
0.1176	0.2526	19.88	19.77	0.11	1.3922	1.0050	107
0.2032	0.3774	21.76	21.68	0.08	1.3187	1.0155	170
0.3404	0.5236	24.01	24.21	−0.20	1.2178	1.0463	240
0.4639	0.6240	25.90	26.05	−0.16	1.1441	1.0913	271
0.5352	0.6746	26.95	26.96	−0.01	1.1086	1.1261	274
0.6668	0.7614	28.33	28.39	−0.06	1.0563	1.2113	249
0.7785	0.8341	29.50	29.39	0.12	1.0252	1.3097	196
0.8455	0.8798	30.09	29.89	0.20	1.0124	1.3829	150
0.9169	0.9320	30.54	30.35	0.19	1.0036	1.4752	88
1.0000	1.0000	30.74	30.74	0.00	1.0000	1.6045	0
$T = 303.15$ K							
0.0000	0.0000	20.43	20.43	0.00	1.5167	1.0000	0
0.0562	0.1365	22.39	22.39	0.00	1.4521	1.0013	56
0.1176	0.2519	24.49	24.30	0.20	1.3881	1.0056	110
0.2032	0.3754	26.69	26.62	0.07	1.3093	1.0169	172
0.3404	0.5205	29.57	29.67	−0.10	1.2057	1.0487	239
0.4639	0.6212	31.67	31.88	−0.21	1.1331	1.0934	267
0.5352	0.6724	32.93	32.99	−0.06	1.0991	1.1272	268
0.6668	0.7609	34.65	34.75	−0.10	1.0504	1.2070	271
0.7785	0.8351	36.25	36.02	0.23	1.0222	1.2957	188
0.8455	0.8814	36.82	36.67	0.15	1.0108	1.3600	143
0.9169	0.9335	37.49	37.28	0.21	1.0031	1.4390	83
1.0000	1.0000	37.84	37.84	0.00	1.0000	1.5467	0
$T = 308.15$ K							
0.0000	0.0000	24.97	24.97	0.00	1.5060	1.0000	0
0.1176	0.2519	29.92	29.70	0.22	1.3892	1.0051	111
0.2032	0.3764	32.50	32.57	−0.07	1.3152	1.0157	174
0.3404	0.5225	36.43	36.38	0.05	1.2144	1.0465	246

Table 5 (continued)

x_1	y_1 calc	P_{exp} (kPa)	P_{calc} (kPa)	$P_e - P_c$ (kPa)	γ_1	γ_2	G^E (J/mol)
$T = 308.15$ K							
0.4639	0.6230	38.87	39.13	−0.26	1.1413	1.0912	277
0.5352	0.6739	40.42	40.50	−0.09	1.1063	1.1257	279
0.6668	0.7611	42.73	42.67	0.06	1.0549	1.2094	254
0.7785	0.8342	44.30	44.18	0.11	1.0245	1.3054	199
0.8455	0.8801	45.09	44.96	0.13	1.0120	1.3765	154
0.9169	0.9323	45.79	45.65	0.14	1.0035	1.4656	90
1.0000	1.0000	46.27	46.27	0.00	1.0000	1.5900	0
$T = 313.15$ K							
0.0000	0.0000	30.36	30.36	0.00	1.5514	1.0000	0
0.1176	0.2535	36.37	36.24	0.13	1.4053	1.0062	118
0.2032	0.3757	39.79	39.70	0.09	1.3184	1.0185	184
0.3404	0.5189	44.01	44.20	−0.19	1.2072	1.0526	255
0.4639	0.6187	47.34	47.46	−0.12	1.1317	1.0993	281
0.5352	0.6699	49.13	49.08	0.05	1.0971	1.1338	281
0.6668	0.7588	51.69	51.67	0.02	1.0485	1.2139	258
0.7785	0.8339	53.48	53.51	−0.03	1.0211	1.3008	194
0.8455	0.8807	54.64	54.48	0.16	1.0102	1.3625	146
0.9169	0.9333	55.49	55.38	0.11	1.0029	1.4371	85
1.0000	1.0000	56.23	56.23	0.00	1.0000	1.5370	0

Redlich–Kister equation with 3 parameters—for mixtures (I) and (II) and 2 parameters—for mixture (III).

Figs. 1–3 illustrate the experimental and calculated isotherms resulting from the same equation; a very good agreement between these data is observed.

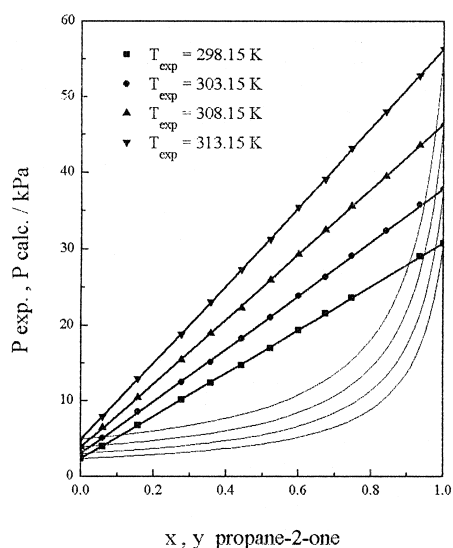


Fig. 2. Experimental and calculated isotherms for propane-2-one + 1,3-dichloropropane mixture.

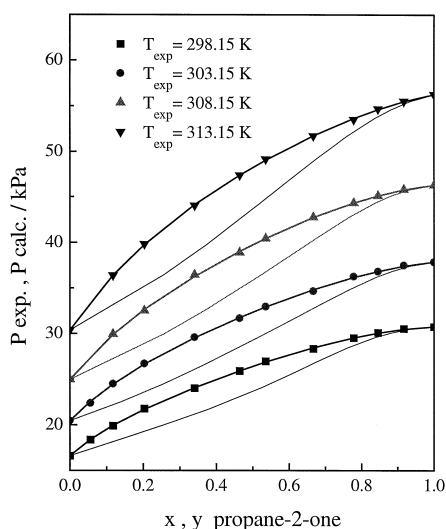


Fig. 3. Experimental and calculated isotherms for propane-2-one + 1,1,1-trichloroethane mixture.

The experimental VLE data were reduced using Barker's method [9]. In this approach, an expression for G^E of the liquid phase is assumed and parameter values in the expression are obtained by regressing $P - x$ data such that the sum of squares between measured and calculated pressures is minimized. The assumed expression for G^E is adequate if the $P - x$ data can be fitted to within experimental uncertainty.

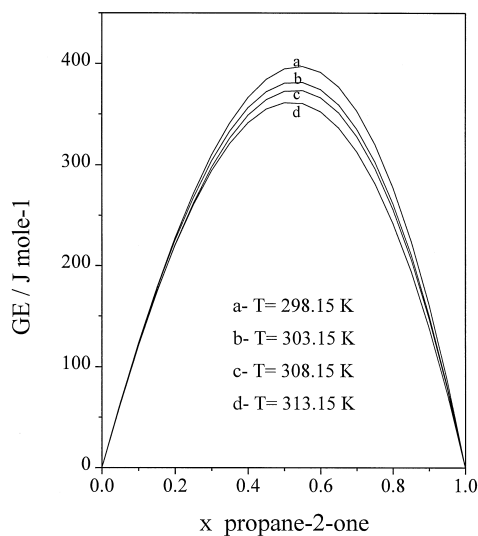


Fig. 4. Molar excess Gibbs energies for propane-2-one + 1-chloropentane mixture.

Table 6

Parameters of the equations used to correlate VLE data for propane-2-one + chloroalkanes mixtures and standard deviation $\sigma(P)$ at 298.15; 303.15; 308.15; 313.15 K

Equation	Parameters ^a				$\sigma(P)/\text{kPa}$			
Propane-2-one + 1-Chloropentane								
<i>Redlich–Kister</i>								
$m = 2$	0.65397	0.62312	0.60280	0.57831	0.18	0.18	0.23	0.10
	0.10656	0.07782	0.07432	0.04670				
$m = 3$	0.65432	0.62307	0.60277	0.57830	0.18	0.14	0.23	0.11
	0.09864	0.06627	0.06847	0.04546				
	0.05881	0.05530	0.02862	0.01009				
<i>Wilson</i>								
$m = 2$	221.4528	193.8109	189.4409	155.3788	0.17	0.15	0.22	0.09
	13.7173	25.6702	25.2686	46.6244				
<i>NRTL $\alpha = 0.3$</i>								
$m = 2$	273.3148	235.3623	231.4299	183.4648	0.18	0.16	0.21	0.10
	−40.3495	−19.3324	−19.6979	13.9095				
Propane-2-one + 1,3-Dichloropropane								
<i>Redlich–Kister</i>								
$m = 2$	−0.02347	−0.03270	−0.03123	−0.04279	0.10	0.12	0.12	0.07
	0.00194	−0.00691	0.00339	−0.03078				
$m = 3$	−0.01478	−0.02536	−0.02506	−0.04288	0.08	0.10	0.08	0.07
	−0.02070	−0.02715	−0.01701	−0.03052				
	0.06014	0.04778	0.04114	−0.00062				
<i>Wilson</i>								
$m = 2$	−152.3229	−194.5162	−169.3188	−226.6427	0.10	0.12	0.31	0.07
	198.7950	280.2666	224.2286	343.3617				
<i>NRTL $\alpha = 0.3$</i>								
$m = 2$	−138.1166	−212.3192	−166.6212	−227.4406	0.10	0.12	0.14	0.06
	153.0331	258.4582	188.2422	363.7268				
Propane-2-one + 1,1,1-Trichloroethane								
<i>Redlich–Kister</i>								
$m = 2$	0.44155	0.42633	0.43659	0.43447	0.14	0.15	0.14	0.11
	0.03126	0.00979	0.02712	−0.00467	0.06	0.09	0.09	0.07
$m = 3$	0.42694	0.41475	0.42934	0.42969				
	0.01989	0.01010	0.01353	−0.01336				
	0.11801	0.09241	0.07542	0.04912				
<i>Wilson</i>								
$m = 2$	124.1383	85.3485	116.3527	64.072	0.14	0.15	0.14	0.11
	20.2745	51.9843	29.5746	80.3603				
<i>NRTL $\alpha = 0.3$</i>								
$m = 2$	149.5643	92.9429	139.4149	58.3125	0.14	0.15	0.14	0.11
	−6.9212	41.3518	4.4276	82.636				

For G^E , the following equations were tested: Redlich–Kister [10], Wilson [11] and NRTL [12]. The parameters of these G^E expressions were calculated by regression, using an objective function of the total pressures:

$$S = \sum_{i=1}^N \left(\frac{P_{i,e} - P_{i,c}}{P_{i,e}} \right)^2 = S(a, b, c, \dots) \quad (3)$$

where: $P_{i,e}$, $P_{i,c}$ stand for the experimental and calculated pressure, N for number of experimental points and a , b , c , ... for the equation parameters.

Vapour phase correction was effected by the two-term virial equation, whose second virial coefficients, for pure compounds as well as mixtures, were calculated from the correlation of Tsonopoulos [13]. The molar volumes were obtained using Rackett's equation whose coefficients were taken from Lide and Kehiaian [14]—for propane-2-one, and estimated by Lydersen's method [15]—for 1-chloropentane, 1,3-dichloropropane and 1,1,1-trichloroethane.

The parameters of the equations used in correlation and the values of standard deviation $\sigma(P)$ are listed in Table 6.

The presented results evidence that the studied mixtures show positive ((I) and (III)) and negative and small ((II)) deviations from ideality. The data can be described by any equation from that used here, with nearly the same accuracy.

Plot of G^E values listed in Tables 3–5, versus the liquid phase composition (Figs. 4–6) show positive maxima at $x \approx 0.5$ and a minimum shifted at $x \approx 0.65$; the maximum values of G^E decreases

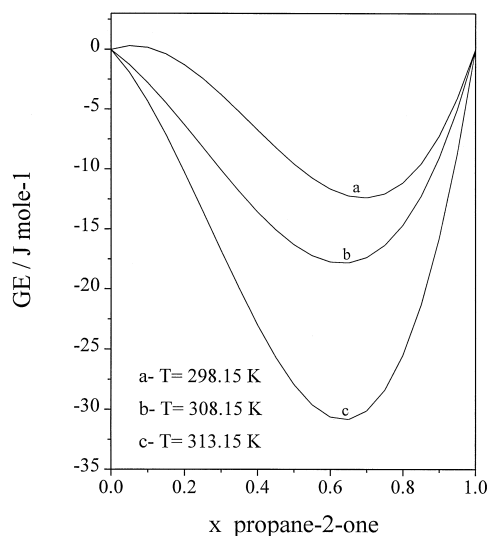


Fig. 5. Molar excess Gibbs energies for propane-2-one + 1,3-dichloropropane mixture.

Notes to Table 6:

^aWilson and NRTL parameters in J/mol.

$\sigma(P)$ is the average standard deviation of the total vapour pressure:

$$\sigma(P) = \left[\sum (P_{\text{exp}} - P_{\text{calc}})^2 / (N - m) \right]^{1/2}.$$

N = number of experimental points; m = number of equation parameters.

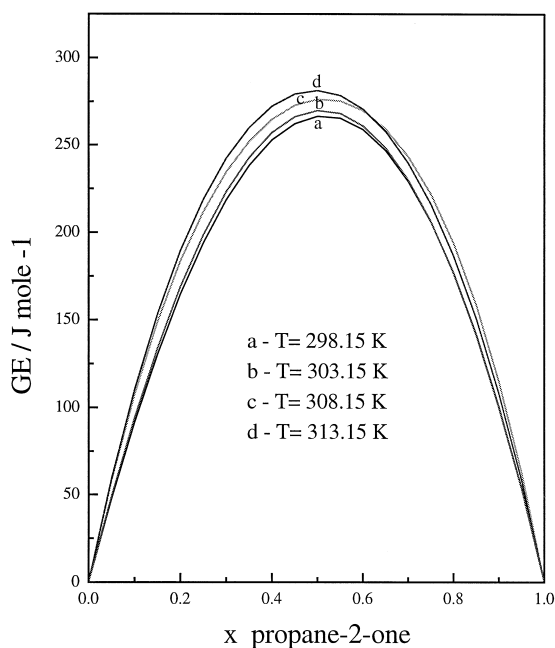


Fig. 6. Molar excess Gibbs energies for propane-2-one + 1,1,1-trichloroethane mixture.

with the temperature increasing in the first and second mixture and increases with the temperature, in the third one.

As can be seen, the effect of the temperature on G^E is very small, thus corresponding to a moderate excess enthalpy.

The equimolar excess enthalpy H^E values, estimated from the temperature dependence of G^E ($x = 0.5$)/ T by the Gibbs–Helmholtz equation, at an average temperature of 305 K are: 976 J/mol, for propane-2-one + 1-chloropentane mixture (I), -330 J/mol, for propane-2-one + 1,3-dichloropropane mixture (II) and 95 J/mol, for propane-2-one + 1,1,1-trichloroethane mixture (III).

For mixture (I) no direct calorimetric H^E data are available, but the value of H^E ($x = 0.5$) = 455 J/mol, at 298.15 K, for propane-2-one + 1-chlorobutane mixture [14] may be considered for a possible comparison.

In the case of mixtures (II) and (III), the calorimetric values of H^E ($x = 0.5$) are: -50 J/mol, at 298.15 K [16] and 228 J/mol, at 363.15 K [17], respectively.

The agreement between the estimated and measured H^E values is reasonable, if the magnification of the errors inherent in the differentiation of the Gibbs–Helmholtz equation and the variation of H^E with the temperature in case of mixture (III) are taken into account.

4. List of symbols

a, b, c	parameters in Eq. (3)
A_0, A_1, A_2, A_3	coefficients of Eq. (1)

G^E	molar excess Gibbs energy, J/mol
m	number of parameters of equations for G^E
$n_D^{298.15\text{ K}}$	refractive index at 298.15 K
n_1, n_2, n	refractive index of pure compounds and of mixture, respectively
N	number of experimental points
P	vapour pressure/kPa
$P_{i,e}, P_{i,c}$	experimental and calculated vapour pressure/kPa
R	general gas constant
S	objective function in Eq. (3)
T	temperature/K
x, y	liquid and vapour mole fraction, respectively
X_{ad}	variable in Eq. (1)
Y	excess function, Eq. (1)
<i>Greek letters</i>	
α	NRTL parameter
γ_1, γ_2	activity coefficients
$\rho_{298.15\text{ K}}$	density of pure compounds/kg/m ³
σ	average standard deviation

Acknowledgements

We express our gratitude to Dr. H.V. Kehiaian (ITODYS, University Paris VII, CNRS, France) for proposing the research program on VLE in ketones + chloroalkanes mixtures and for comments and helpful discussions.

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